An Integrated Meta-Modeling and Generative-AI Approach to Identify Dust Prediction Errors Incorporating Online Repositories and Climate Change Projections

By
Ibrahim Deeb Kukash

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Supervisor: Dr. Mohamed Abdallah

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Abstract

This research proposes a meta-modelling approach that shifts the focus from predicting dust directly to predicting model failure. The framework will treat forecast verification as a classification problem, analyzing historical forecast-observation discrepancies from local monitoring stations. Machine learning classifiers, including Random Forest, XGBoost, and Neural Networks, will be trained on ERA5 meteorological features to detect multi-source predictive signatures with high error rates. The study will evaluate both standard physics-based models and emerging global AI weather emulators to characterize distinct failure modes.

Current forecasting methods generally fail to account for human-driven changes that standard monitoring techniques may not capture. To address this, the proposed framework integrates a novel data stream: generative AI agents deployed to mine historical documents for events such as agricultural collapse, marsh drainage, and water policy shifts. These features will be geocoded and tested to quantify their contribution to prediction accuracy.

Finally, the research will assess model stability under non-stationary climate conditions. By applying the trained error classifiers to CMIP6 projections (SSP2-4.5 and SSP5-8.5), the study intends to estimate failure trends through 2050. This aims to identify specific temperature and precipitation thresholds that may necessitate fundamental model recalibration. The expected outcome is a set of operational tools for climate-adaptive forecast evaluation, demonstrating that meta-modeling can serve as an effective method for uncertainty quantification.

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Chapter 1: Introduction

1.1 Background

The Arabian Gulf is one of the most dust-active regions globally, where frequent sand and dust storms (SDS) pose severe risks to public health, aviation safety, solar energy generation, and economic infrastructure. Accurate forecasting of these events is critical for mitigation and operational planning. However, despite decades of development in Numerical Weather Prediction (NWP), predicting the onset, intensity, and transport of dust in this region remains a formidable challenge.

The region's unique topography and meteorology create complex dust dynamics. The dominant "Shamal" winds, driven by pressure gradients between the Saudi thermal low and the Zagros Mountains, mobilize vast quantities of mineral dust from the Mesopotamian floodplain and the Rub' al Khali desert. These synoptic-scale events are often compounded by mesoscale convective systems (haboobs), which occur on spatial scales too small for many global models to resolve effectively. As a result, operational systems often struggle to capture the precise timing and magnitude of dust outbreaks, leading to significant forecast errors.

These modeling difficulties are exacerbated by the rapid and ongoing transformation of the regional environment. Anthropogenic activities—specifically the drainage of Mesopotamian marshes, the collapse of agriculture across the Fertile Crescent due to drought and conflict, and rapid urbanization—have created new, potent dust sources. These "anthropogenic dust sources" were largely absent or dormant when current operational forecasting systems were initially parameterized. Consequently, both physics-based models (which rely on static source inventories) and emerging AI-based models (which learn patterns from historical reanalysis data) suffer from a "blind spot" that fails to account for land-surface changes that fundamentally alter emission potentials.

1.2 Problem Identification

Although physics-based and AI models exist independently, the literature lacks a direct, head-to-head evaluation of these systems over the Arabian Gulf. The atmospheric science community has extensive experience benchmarking different physics-based models, yet the AI community's parallel advancements remain largely separate. Consequently, it is unknown whether CAMS and emerging AI aerosol emulators make similar errors under similar conditions (suggesting fundamental limits in understanding dust processes) or whether they fail in complementary ways. This knowledge gap prevents informed decisions about which modeling concepts to invest in for operational forecasting infrastructure.

1.3 Research Objectives

This research addresses the identified gaps through an integrated analysis that combines systematic benchmarking, meta-model development, and climate-robustness assessment. The specific objectives are:

Objective 1: Systematic Benchmarking of Physics-Based and AI-Based Dust Forecasting Systems
Conduct performance evaluation of CAMS and emerging AI weather emulators for dust prediction over the Arabian Gulf, using historical forecast-observation pairs from local monitoring stations. The analysis will identify distinct failure modes for each modeling approach, determine whether errors are correlated or complementary, and characterize atmospheric regimes where each system demonstrates superior skill.

Objective 2: Development of a Meta-Modeling Framework for Forecast Failure Prediction
Develop and validate machine learning classifiers (Random Forest, XGBoost, Neural Networks) that predict when dust forecasts from physics-based and AI-based systems are likely to exhibit high error rates. The meta-model will use ERA5 meteorological features, satellite-derived indicators of land surface change, and temporal trend features to enable conditional confidence assessment for operational forecasting.

Objective 3: Integration of Anthropogenic Dust Source Indicators via Generative AI
Deploy generative AI agents to mine historical documents for evidence of anthropogenic events affecting dust sources (agricultural collapse, marsh drainage, water policy shifts, conflict-related land disturbance). Geocode extracted events and test their contribution to meta-model prediction accuracy, quantifying whether anthropogenic source changes explain forecast failures that meteorological features alone cannot capture.

Objective 4: Assessment of Forecasting System Robustness Under Climate Change Scenarios
Apply trained error classifiers to CMIP6-projected meteorology (SSP2-4.5 and SSP5-8.5 scenarios) to estimate how forecast failure rates may change through 2050. Identify specific temperature and precipitation thresholds that may necessitate model recalibration, and compare robustness of physics-based versus AI-based approaches to climate-driven distribution shifts.

Objective 5: Formulation of Operational Guidelines
Synthesize findings into actionable recommendations for operational forecasting centers, including confidence estimation workflows, dynamic model selection strategies, and climate-adaptive forecast evaluation protocols demonstrating meta-modeling as an effective uncertainty quantification method.

1.4 Scope of Work

The study focuses on the Arabian Gulf region, encompassing the United Arab Emirates, Qatar, Bahrain, Kuwait, eastern Saudi Arabia, and southern Iraq. This domain captures major dust source regions (Mesopotamian floodplain, dried marshlands, degraded agricultural areas) and affected population centers.

The climatological analysis of CAMS performance will cover 2015-2024, capturing the whole operational period and the availability of high-quality ERA5 reanalysis (Hersbach et al., 2020). Comparative benchmarking of AI-based systems will focus on the post-2021 period to ensure rigorous out-of-sample evaluation and avoid data leakage from model training periods. Future projections will extend to 2050 using CMIP6 scenarios.

The research will focus on CAMS as the primary physics-based operational system and will evaluate available global AI weather emulators that include aerosol predictions. Specific AI systems will be selected based on data accessibility, reproducibility, and documented performance for aerosol variables.

Ground-based observations from regional air quality monitoring networks, AERONET sun photometer measurements, MODIS and VIIRS satellite aerosol retrievals, ERA5 reanalysis meteorological fields, CMIP6 climate projection outputs, and historical document archives for anthropogenic event extraction.

Meta-model development will employ established supervised learning algorithms (Random Forest, XGBoost, Neural Networks) selected for their data efficiency and interpretability. Generative AI for document mining will use state-of-the-art large language models with appropriate safeguards against hallucination.

Limitations:
The study acknowledges data sparsity in ground-truth dust observations, challenges in isolating anthropogenic versus natural source changes, uncertainties in CMIP6 projections, and the potential for AI model access restrictions that may limit which systems can be evaluated.

1.5 Significance and Contribution

This research bridges the gap between traditional atmospheric physics and emerging artificial intelligence, offering direct contributions to science, operations, and regional planning.

Scientific Contributions First, this study provides the first head-to-head audit of physics-based versus AI-driven dust forecasting in the Gulf. We determine if AI models offer true

predictive gains or achieve computational speed while replicating the physical biases of their training data. Second, the proposed framework treats forecast error as a predictable variable. This shifts the field from static bias correction to dynamic confidence estimation—allowing us to predict *when* a model will fail, not just *that* it failed. Finally, the use of Large Language Models (LLMs) to digitize qualitative historical records (like marsh drainage reports) creates a reproducible workflow for integrating unstructured data into quantitative environmental models.

Operational Contributions. For forecasting centres, this research delivers a "warning system" for model reliability. Instead of treating every forecast as equally valid, operators will gain tools to identify atmospheric regimes where specific models are untrustworthy. Additionally, a climate robustness assessment provides a timeline for the system's lifespan. By identifying the specific temperature or precipitation thresholds that degrade forecast skill, agencies can proactively plan model recalibrations rather than react to future failures.

Broader Implications: While focused on the Arabian Gulf, the methodology is universal. The framework for stress-testing AI models against climate change and the technique for mining anthropogenic history can be applied to any environmental system facing non-stationary conditions, from hydrology to urban heat monitoring.

1.6 Work Plan

The research is scheduled for the Spring 2026 semester (January–May) and is divided into five phases to ensure the timely completion of all objectives. The timeline and dependencies are shown in Figure 1.1.

Phase 1: Data Acquisition & Baseline Benchmarking (January) This phase involves collecting and processing the core datasets: CAMS historical forecasts (2015–2024), ERA5 reanalysis, and AERONET ground observations. A baseline analysis will calculate standard error metrics (RMSE, bias) for the physics-based models to establish a reference point for evaluating the meta-model.

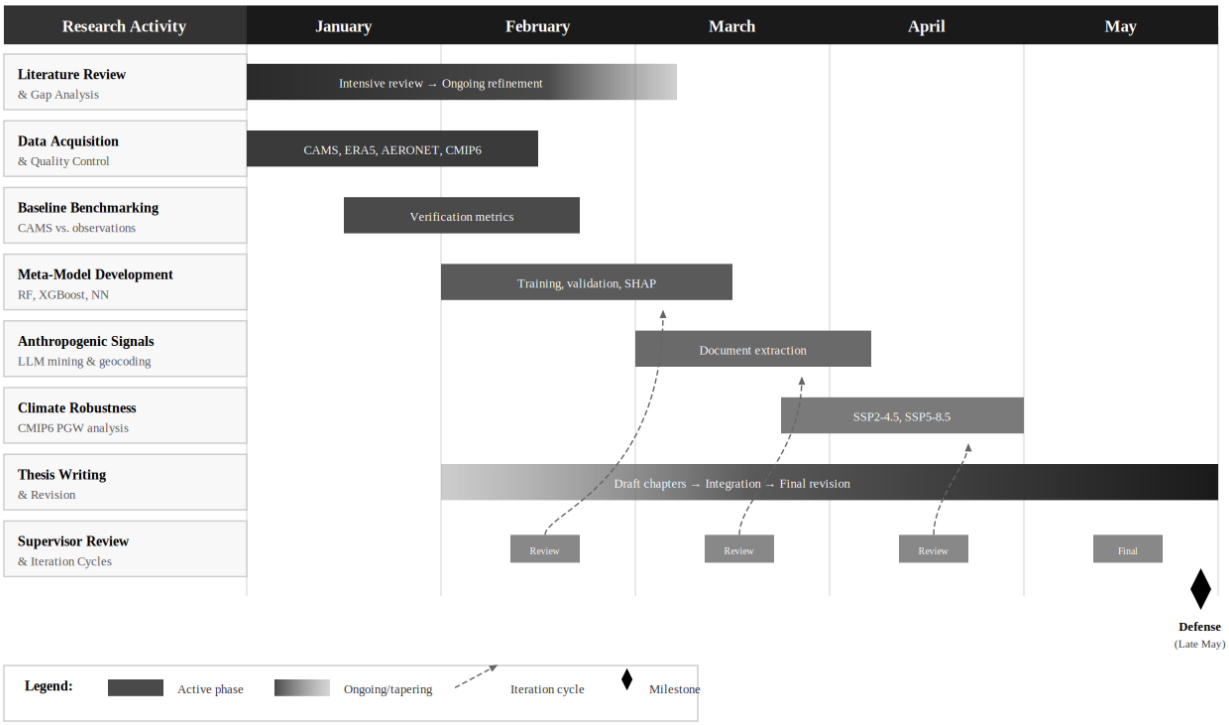
Phase 2: Meta-Model Training & Validation (February) Machine learning classifiers (Random Forest, XGBoost, Neural Networks) will be trained to predict forecast errors using meteorological predictors. The models will undergo temporal cross-validation to prevent data leakage. SHAP values will be computed to identify which atmospheric variables contribute most to forecast failures.

Phase 3: Anthropogenic Signal Integration (March) Generative AI tools will be used to extract records of marsh drainage and agricultural abandonment from historical documents. These events will be geocoded and added as binary features to the meta-model. The analysis will then quantify whether adding these anthropogenic indicators improves the model's ability to predict failures.

Phase 4: Climate Robustness Assessment (April) The trained meta-models will be tested against future climate conditions using CMIP6 projections (SSP2-4.5 and SSP5-8.5). This phase estimates how failure rates might change through 2050 and identifies specific meteorological thresholds that could degrade model performance.

Phase 5: Synthesis & Defense (May) The final phase focuses on interpreting the results and formulating operational guidelines. Work includes finalizing the thesis document, incorporating supervisor feedback, and preparing the defense presentation.

Figure 1.1: Research Timeline (Spring 2026 Semester)



1.7 Thesis Organization

The thesis is organized into the following chapters:

Chapter 1: Introduction provides background on dust-forecasting challenges in the Arabian Gulf, identifies specific research gaps, states the research objectives, defines the scope of work, and outlines the significance of the contributions.

Chapter 2: Literature Review examines the current state of operational dust forecasting systems (CAMS, DREAM8b, emerging AI emulators), reviews machine learning approaches to forecast error prediction and meta-modeling, discusses anthropogenic drivers that have altered dust source geography, explores climate change impacts on forecasting system reliability, and synthesizes these elements to justify the research approach.

Chapter 3: Research Methodology details the data sources and quality control procedures, describes the systematic benchmarking framework for comparing physics-based and AI-based systems, explains the meta-model development process, including algorithm selection and feature engineering, outlines the generative AI approach for anthropogenic signal extraction, and specifies the climate robustness testing methodology using CMIP6 projections.

Chapter 2: Literature Review

2.1 Overview

The review is organized as follows: Section 2.2 examines operational dust forecasting systems currently deployed over the Arabian Gulf, with emphasis on their capabilities and documented limitations. Section 2.3 explores emerging machine learning techniques for predicting when numerical models are likely to fail. Section 2.4 discusses anthropogenic factors that have altered dust source geography in recent decades. Section 2.5 examines how climate change may affect forecasting system reliability over coming decades. Finally, Section 2.6 synthesizes these elements to identify specific research gaps.

2.2 Operational Dust Forecasting Systems

Operational dust forecasting relies predominantly on physics-based numerical models that simulate atmospheric dynamics, aerosol transport, and surface-atmosphere interactions. Two primary systems serve the Arabian Gulf region: the Copernicus Atmosphere Monitoring Service (CAMS) and the Barcelona Supercomputing Center's DREAM8b model. Both systems have been extensively validated but exhibit performance variability that depends on meteorological conditions and dust source characteristics. A detailed comparison of physics-based (CAMS) versus AI-based forecasting approaches across multiple dimensions is provided in Table 2.1.

Table 2.1: Comparison of Dust Forecasting Approaches

Dimension	Physics-Based (CAMS)	AI-Based Emulators
Computational Cost	High (hours on supercomputers)	Low (minutes on GPU)
Physical Consistency	Conservation laws enforced	Learned patterns; may violate physics
Spatial Resolution	~40 km (limited by computation)	~25 km (limited by training data)
Extreme Events	Underestimates peaks (resolution)	Underestimates peaks (MSE training)
Source Representation	Static inventory (may be outdated)	Implicit in training data patterns
Climate Robustness	Parameterizations may drift	Distribution shift vulnerability
Interpretability	Process-based (traceable)	Black box (requires XAI methods)

2.2.1 CAMS Global Forecasting System

The Copernicus Atmosphere Monitoring Service provides global aerosol forecasts updated twice daily at 00:00 and 12:00 UTC, delivering five-day predictions at approximately 40 km horizontal resolution (ECMWF, 2024). CAMS uses the Integrated Forecasting System coupled with aerosol modules that simulate dust emission, transport, and deposition. Dust emissions are parameterized based on surface wind speed, soil moisture, vegetation cover, and a static inventory of erodible surfaces compiled from satellite observations.

It is important to note that CAMS is an evolving operational system, meaning its underlying physics and data assimilation schemes change over time. For instance, the September 2017 operational upgrade (Figure 2.2) introduced interactive aerosol-chemistry schemes and assimilated new PMAp satellite data (CAMS, 2017). These continuous upgrades imply that the model's error characteristics are non-stationary, further motivating the need for dynamic meta-modeling rather than static bias correction.

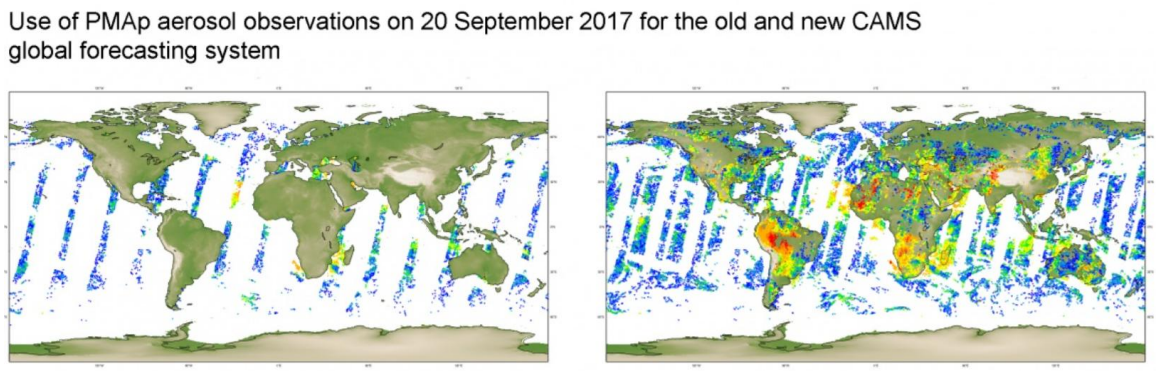


Figure 2.2: Impact of the September 2017 CAMS operational upgrade on forecast accuracy. ***Source: Copernicus Atmosphere Monitoring Service (CAMS)/ECMWF (2017).***

Validation studies demonstrate that CAMS performs reasonably well for large-scale dust transport but encounters difficulties with event timing and intensity in source regions. Akritidis et al. (2020) evaluated CAMS against AERONET sun photometer observations across the Mediterranean and Middle East, finding that the model tends to underestimate aerosol optical depth during extreme dust episodes while overestimating background conditions. More recent analysis over the Eastern Mediterranean showed correlation coefficients ranging from 0.57 to 0.85 across different stations, with mean absolute errors of 0.08 and root mean square errors of 0.11 (Tuna Tuygun & Elbir, 2024).

For the Arabian Peninsula specifically, CAMS exhibits systematic biases. The model consistently underperforms during convectively-driven dust events (haboobs) where cold pool winds from decaying thunderstorms mobilize dust in otherwise stable atmospheric conditions. These events develop at spatial scales of 10-20 km, which are unresolved by

CAMS's 40 km grid. CAMS also relies on dust source inventories that do not account for recent anthropogenic changes, such as marsh drainage in Iraq or agricultural collapse in Syria, which may contribute to spatial mismatches between predicted and observed dust plumes.

2.2.2 Regional Dust Modeling: BSC-DREAM8b

The Barcelona Supercomputing Center's DREAM8b model provides regional dust forecasts for North Africa, the Mediterranean, and the Middle East at 0.3° resolution (approximately 30 km). Unlike CAMS's global approach, DREAM8b specializes in mineral dust. It uses eight size bins to represent the particle size distribution, enabling more accurate simulation of differential transport between fine- and coarse-dust fractions (Basart et al., 2012).

DREAM8b has been operational since 2006, driven by meteorological forecasts from NCEP's Global Forecast System. The model incorporates dust emission parameterizations that account for soil texture, surface roughness, and soil moisture, with representation of sub-grid scale wind variability to capture dust mobilization by turbulent gusts. Validation against AERONET observations shows reasonable skill for moderate dust events but underestimation of peak aerosol optical depth during extreme Saharan outbreaks (Konsta et al., 2018).

In the Arabian Gulf, DREAM8b faces similar challenges to CAMS. Evaluation against AERONET stations in the UAE and Saudi Arabia revealed that while the model captures seasonal dust activity patterns, it consistently misses short-duration events associated with convective activity and tends to overpredict dust during winter when actual dust loads are low (Karami et al., 2021). These systematic errors suggest fundamental limitations in how both physics-based systems represent the interaction between meteorology and dust emission processes in this region.

2.2.3 Emerging AI-Based Weather Prediction Systems

The past two years have witnessed dramatic advances in machine learning-based weather prediction (Figure 2.1 illustrates this evolution). Systems such as GraphCast, Pangu-Weather, FuXi, and GenCast have demonstrated that data-driven models can match or exceed traditional physics-based forecasts for general meteorological variables (Lam et al., 2023; Bi et al., 2023; Chen et al., 2023; Price et al., 2024). These models learn spatiotemporal patterns from decades of reanalysis data, using architecture based on transformers, graph neural networks, or diffusion models.

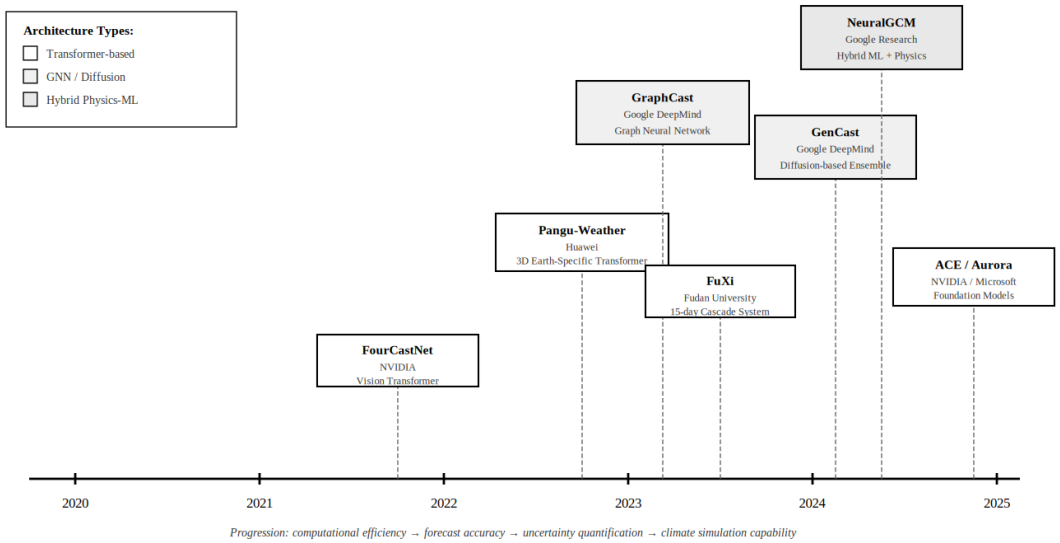


Figure 2.1: Chronological evolution of major AI-based weather prediction architectures (2020–2025).

Unlike physics-based systems that explicitly solve atmospheric equations, AI weather models learn statistical relationships between atmospheric states and their temporal evolution. This approach offers computational advantages—generating global forecasts in minutes rather than hours—but raises questions about extrapolation beyond training data and physical consistency (Bonavita, 2024). Recent comparative assessments show that while these models achieve competitive skill for standard meteorological variables, they tend to underpredict extremes due to training objectives that minimize mean squared error, resulting in smoother forecasts than physics-based ensembles (Schultz et al., 2024).

The extension of AI weather prediction to aerosol forecasting remains in the early stages. While some global AI weather models include aerosol variables in their outputs, systematic validation of dust predictions specifically over the Arabian Gulf has not been published. The fundamental question of whether AI models trained on historical reanalysis reproduce the same biases as the underlying physics models, or develop distinct failure modes, remains unanswered. This uncertainty is particularly relevant for regions like the Arabian Gulf, where dust source geography has changed substantially over the training period used by most AI systems.

In addition to global academic initiatives like AI-GAMFS, commercial partnerships are rapidly emerging in the region. Notably, the collaboration between G42 and NVIDIA (announced in 2024) uses the Earth-2 platform and CorrDiff generative AI to produce hyper-local dust forecasts for the UAE. While currently proprietary and unavailable for public benchmarking, this initiative highlights the operational urgency of shifting from physics-based to AI-driven aerosol forecasting in the Arabian Gulf.

The field is moving beyond short-term weather prediction toward climate-capable AI systems. Recent developments include the Ai2 Climate Emulator (ACE) from NVIDIA and the Allen Institute, which uses generative conditional sampling to produce ensemble climate projections from ERA5 reanalysis (Brenowitz et al., 2025). Google Research's NeuralGCM demonstrates hybrid architectures combining machine-learned physics with differentiable atmospheric cores, achieving accurate weather forecasts while maintaining physical consistency for multi-decadal climate simulations (Kochkov et al., 2024). The University of Washington's DLESyM extends AI forecasting to include ocean coupling, enabling the reproduction of stable seasonal cycles and the simulation of ENSO variability (Cresswell-Clay et al., 2025). These systems represent a fundamental shift from pattern recognition trained on historical weather toward AI models capable of simulating climate under novel forcing conditions. However, research on whether these capabilities extend to regional aerosol processes under changing climate conditions remains absent from the literature.

2.3 Machine Learning for Forecast Error Prediction

Beyond directly generating forecasts with machine learning, an alternative approach uses it to predict when and why conventional forecasting systems fail. This meta-modeling paradigm treats forecast error as a predictable quantity that can be estimated from atmospheric conditions and model characteristics, enabling conditional confidence assessment rather than uniform bias correction.

2.3.1 The Meta-Modeling Framework

Traditional forecast post-processing typically applies static bias corrections based on long-term mean errors. Model Output Statistics (MOS), used operationally since the 1970s, adjusts temperature and precipitation forecasts using historical model performance at specific locations. While effective at addressing systematic biases, these methods struggle with conditional failures, in which models perform well under specific atmospheric regimes but catastrophically fail under others.

Meta-modeling shifts focus from correcting forecasts to predicting forecast reliability. Rather than asking "what is the correct forecast?", meta-models ask "how wrong is this forecast likely to be given current conditions?" This enables forecasters to identify when model outputs are unreliable before decisions are made. For example, a meta-model might recognize that under high atmospheric stability with weak winds, a dust model slightly overpredicts aerosol loads, while under convective conditions with strong vertical wind shear, the same model severely underpredicts dust mobilization.

Research in this direction remains limited. Most applications have focused on precipitation and temperature forecasting rather than aerosol prediction. The challenges include data scarcity (sufficient forecast-observation pairs covering diverse atmospheric

conditions), multiple sources of error (meteorology, emissions, transport, deposition), and non-stationarity (relationships between conditions and errors change as models are updated or the climate evolves).

2.3.2 Applications for Weather Prediction

Few studies have systematically applied meta-modeling to atmospheric predictions. Moosavi et al. (2013) demonstrated the concept for precipitation forecasts from the Weather Research and Forecasting model, using Random Forest and Neural Networks to learn relationships between model parameterization choices and resulting forecast errors. Their key finding was that optimal parameterization schemes vary with atmospheric regime—no single configuration performs best under all conditions. By training on historical forecast-observation pairs, they could predict which scheme combinations would minimize errors for specific synoptic setups.

More recently, efforts have focused on using machine learning to estimate model error in data assimilation systems. These techniques identify systematic biases in background forecasts that can be corrected before assimilating observations, potentially improving both analysis quality and subsequent forecast skill (Laloyaux et al., 2024). However, these approaches target analysis error rather than predicting forecast failure.

For aerosol forecasting specifically, no published studies have developed meta-models to predict when dust forecasts are unreliable. The concept remains unexplored despite clear operational value. If forecasters knew in advance that atmospheric conditions favor model failure, they could issue hedged predictions, seek alternative information sources, or adjust warning thresholds. This gap is particularly acute for regions experiencing rapid environmental change where model assumptions may become increasingly violated.

2.4 Anthropogenic Drivers of Dust Emissions

While climate variability has always controlled dust activity, human activities over the past decades have fundamentally altered the geography of dust sources in the Middle East. Wetland drainage, agricultural collapse, armed conflict, and urbanization have each created new dust sources that were absent when current operational models were parameterized. These changes pose challenges for both physics-based models (which use static source inventories) and AI models (which learn from historical data that may not reflect current conditions).

2.4.1 Mesopotamian Marsh Drainage

The systematic drainage of Mesopotamian marshes in southern Iraq between the 1990s and early 2000s exposed thousands of square kilometers of fine-grained sediments. These marshes had historically acted as dust sinks, trapping particles and maintaining high soil moisture, thereby preventing wind erosion. Ginoux et al. (2012) identified dried marsh

beds as major new dust sources detectable in satellite imagery, with dust plumes from former wetland areas becoming increasingly common after 2000.

The timing of this transformation is significant for forecasting models. Physics-based systems typically use land surface datasets compiled in the early 2000s to define potential dust sources. Some models may still represent former marsh areas based on pre-drainage conditions, even though current surfaces consist of exposed sediment with negligible vegetation. Partial reflooding efforts after 2003 restored some areas, but large sections remain dry and continue to emit dust, particularly during spring and summer when evaporation keeps soil moisture low (Al-Dousari et al., 2017).

2.4.2 Agricultural Abandonment in the Fertile Crescent

The Fertile Crescent region spanning Syria, northern Iraq, and southeastern Turkey experienced severe multi-year drought from 2007 to 2010, coinciding with political instability that disrupted agricultural practices (Kelley et al., 2015). The combination of natural drought and socioeconomic collapse led to widespread agricultural abandonment. Fields that had been irrigated and cultivated for decades reverted to bare soil, with crop residues removed and no vegetation regrowth.

Satellite observations tracked the expansion of dust-emitting surfaces, showing that areas classified as cropland in 2005 became major dust sources by 2015 (Lee et al., 2025). The transition was not simply a return to natural conditions—abandoned agricultural land has different characteristics than undisturbed desert. Decades of plowing and irrigation alter soil structure, creating fine-grained, loosely aggregated material that is easily mobilized by wind. Additionally, vegetation removal during abandonment eliminates root networks and surface roughness that would normally stabilize soil.

This agricultural collapse creates problems for forecasting models that assume land use is relatively stable. While CAMS updates vegetation coverage seasonally based on satellite greenness indices, its definition of potential dust source regions is static. It does not account for the fact that former cropland can become more erodible than surrounding natural desert.

2.5 Climate Change and Model Non-Stationarity

Dust forecasting models are fundamentally calibrated on historical relationships between atmospheric conditions and dust emissions. Climate change is altering these relationships in ways that could degrade model performance over time. The Arabian Gulf region is particularly vulnerable, with climate projections suggesting substantial changes in meteorological conditions that control dust activity by mid-century.

2.5.1 Projected Climate Changes in the Arabian Gulf

Analysis of 31 CMIP6 models shows that the Arabian Peninsula is expected to warm faster than the global average, with temperature increases of 2-4°C by 2050 under moderate emission scenarios (SSP2-4.5) and 4-6°C by 2100 under high-emission scenarios (SSP5-8.5) (Almazroui et al., 2020). Changes in precipitation patterns and atmospheric circulation accompany this warming.

More critically for dust forecasting, projected changes include modifications to wind patterns and atmospheric stability. The northern Arabian Gulf experiences dust mobilization primarily through Shamal wind events—northwesterly winds associated with synoptic-scale pressure gradients. CMIP6 models project weakening of the pressure gradient between the Saudi thermal low and the Zagros high pressure, the primary driver of summer Shamal events. Some models also show shifts in the subtropical jet stream's position and intensity, which influence spring Shamal frequency, though considerable inter-model spread exists in these projections.

Temperature increases affect dust emissions through multiple pathways. Higher surface temperatures alter boundary layer structure, potentially changing threshold wind speeds required for dust mobilization even if mean wind speeds remain constant. Almazroui et al. (2025) examined aridity projections. They found that most of the Arabian Peninsula will experience increases in aridity index, with largest changes in currently marginal zones between true desert and semi-arid regions.

2.5.2 Implications for Forecasting System Reliability

These projected climate changes create a fundamental problem: forecasting systems calibrated on historical periods (roughly 1980-2020) may not perform reliably under future climate conditions. This non-stationarity problem violates the assumption that past performance indicates future performance.

For physics-based models, non-stationarity manifests through parameterizations. Dust emission schemes use empirical relationships derived from field measurements and wind tunnel experiments conducted under present-day conditions. If relationships between wind turbulence and dust flux change under warmer, drier conditions due to altered boundary layer structure, parameterizations become incorrect. The model might correctly predict wind speed but still miscalculate emission flux because underlying physics have shifted.

AI-based models face even more severe non-stationarity challenges. Systems trained on historical reanalysis data learned patterns like "when the synoptic setup looks like X, dust optical depth will be Y." If climate change alters relationships between synoptic patterns and dust outcomes, these learned associations become unreliable. This is a form of

distribution shift—test data (future climate) comes from a different statistical distribution than training data (historical climate). Machine learning models generally struggle with distribution shift unless specifically designed to handle it through techniques like domain adaptation or continual learning.

Testing model robustness under projected future conditions could identify which forecasting approaches are more vulnerable to climate-driven degradation. This information is valuable for operational planning and for understanding whether physics-based process understanding or pattern recognition approaches offer better long-term reliability.

2.6 Research Gaps and Synthesis

The preceding review reveals that operational dust forecasting faces converging challenges. Physics-based systems struggle with spatial resolution limitations, static dust source inventories, and parameterizations calibrated on conditions that may no longer represent reality. Emerging AI-based weather prediction systems offer computational efficiency and pattern-recognition capabilities but inherit biases from their training data and lack mechanisms to handle distribution shifts when conditions diverge from historical norms. Both approaches face potential performance degradation due to anthropogenic activities altering the geography of dust sources and climate change modifying the meteorological drivers of dust emission and transport.

Despite extensive research on individual forecasting systems and on specific aspects of dust phenomenology, several critical gaps remain. First, no studies have systematically compared the performance of physics-based and AI-based dust forecasting systems in head-to-head evaluations over the Arabian Gulf. While the atmospheric science community has experience benchmarking different physics-based models, and the AI community has demonstrated machine learning potential for weather prediction, these research streams have mainly remained separate. Understanding whether different modeling approaches make similar errors under similar conditions, or whether they fail in complementary ways, remains unknown.

Second, while meta-modeling concepts have been explored for precipitation and temperature forecasts, they have not been applied to aerosol prediction systems. The challenges of data scarcity, multiple error sources, and non-stationarity have deterred research in this direction, yet these challenges precisely motivate the need for conditional failure prediction. Dust forecasting operates in a data-sparse, rapidly-evolving environment where simple bias correction is insufficient and where understanding specific atmospheric regimes that trigger failures could substantially improve operational decision-making.

Third, existing validation studies treat model performance as stationary, evaluating skill over historical periods without considering how performance might change under future conditions. CMIP6 climate projections indicate that the Arabian Gulf meteorological regime will shift substantially by mid-century, yet no research has assessed the robustness of operational forecasting systems under these projected conditions. This omission is critical for systems intended to provide reliable forecasts over multi-decadal planning horizons relevant to infrastructure development, renewable energy investments, and public health preparedness.

These gaps are interconnected. Without understanding distinct failure modes of different modeling approaches, forecasters cannot determine which systems are more vulnerable to anthropogenic source changes or climate shifts. Without frameworks for predicting when forecasts are unreliable, operational users lack guidance on confidence levels. Without testing model robustness under future scenarios, planners cannot assess whether current systems will remain fit-for-purpose or whether fundamental redesign is needed. The Arabian Gulf presents a particularly acute manifestation of these challenges—a region where anthropogenic dust sources have emerged rapidly, where climate change impacts are projected to be severe, and where forecast users require high-confidence predictions for high-stakes decisions.

Chapter 3: Research Methodology

3.1 Overview

This chapter details the analytical framework used to evaluate dust forecasting systems and develop a failure-prediction meta-model. The methodology follows a sequential workflow: data acquisition and quality control, systematic benchmarking of physics-based versus AI-driven models, training of error-prediction classifiers, integration of anthropogenic source indicators, and stress-testing under future climate scenarios. The complete research framework is illustrated in Figure 3.1.

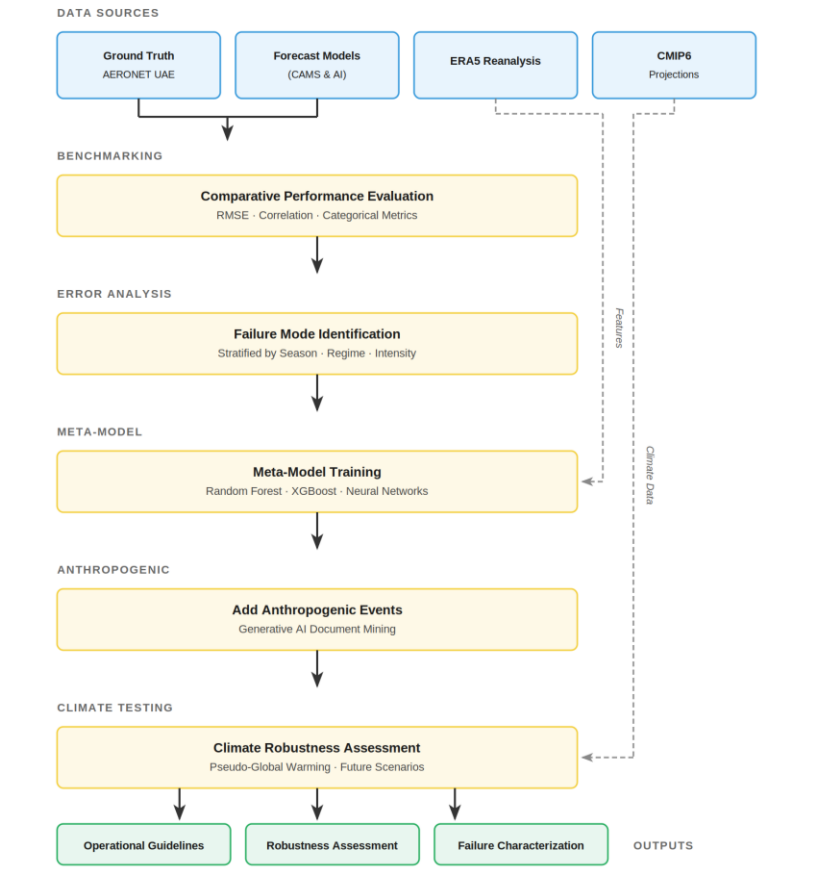


Figure 3.1: Research Methodology Framework illustrating the integrated workflow from data acquisition through comparative benchmarking, meta-model development, and climate robustness assessment.

3.2 Data Sources and Quality Control

This research integrates observational data, numerical forecasts, reanalysis fields, and climate projections. To ensure reliability, all datasets undergo strict quality control, including the removal of physically implausible values and temporal consistency checks.

The study focuses on the Arabian Gulf domain (45°E–60°E, 20°N–32°N), a region characterized by complex interactions between natural desert sources and anthropogenic

land degradation. The spatial extent of this domain, along with major dust source regions and prevailing transport pathways, is illustrated in Figure 3.2.

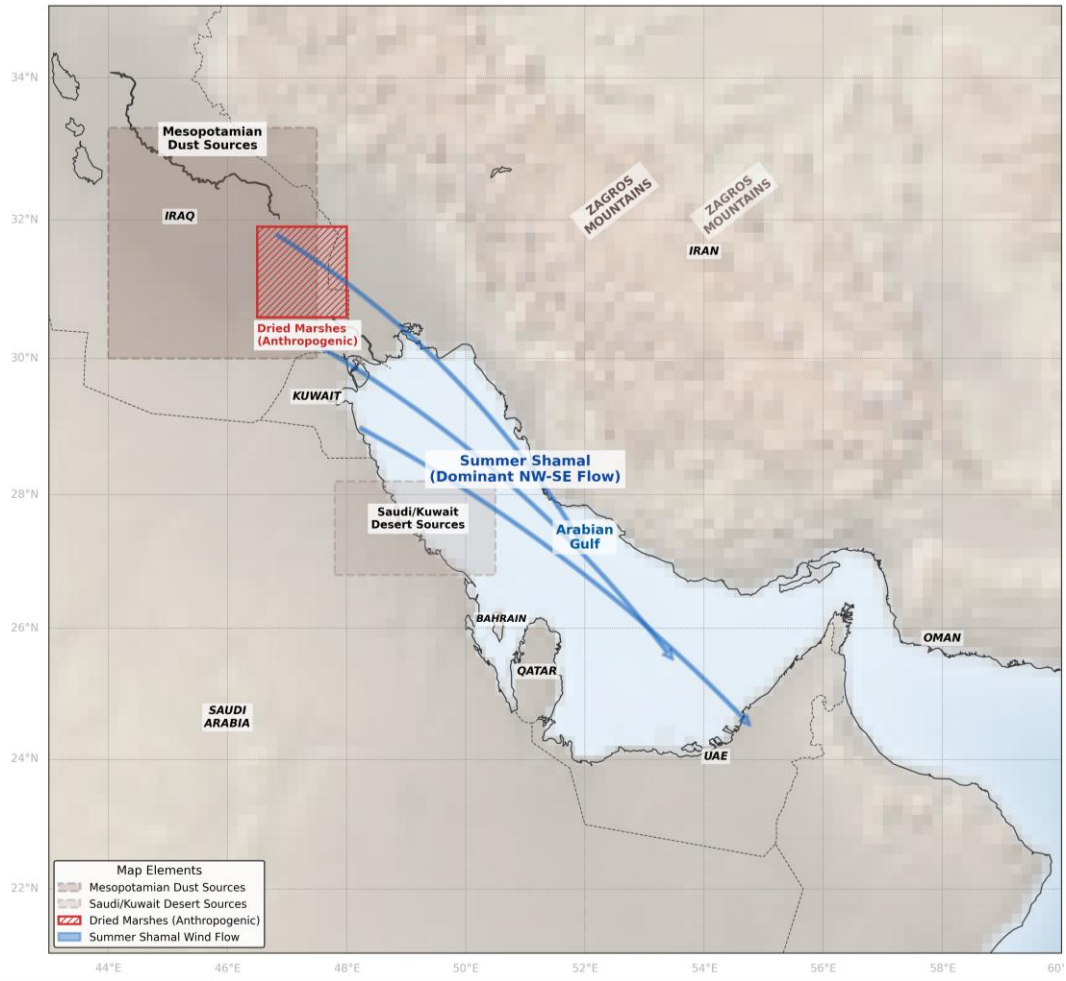


Figure 3.2: Map of the study domain (45°E–60°E, 20°N–34°N) illustrating major dust source regions, including the anthropogenic dried marshes in southern Iraq, and the dominant summer Shamal wind flow (NW-SE) responsible for transporting dust into the Arabian Gulf.

A summary of the datasets, their specifications, and their specific application in this research is provided in Table 3.1.

Table 3.1: Summary of Research Datasets

Dataset Category	Source / Instrument	Resolution & Period	Application in Research
Ground Truth	AERONET (Sun Photometers) & UAE Air Quality Stations (PM10/2.5)	Point-source; High-frequency (15-min to hourly).	Serves as the primary verification for benchmarking CAMS and AI models.

Forecast Models	CAMS Global Atmospheric Composition Forecasts	0.4° spatial; Daily initialization (00 UTC); 2015–2024.	The physics-based baseline model being evaluated.
AI Models	Global AI Weather Emulators (e.g., GraphCast, Pangu-Weather)	0.25° spatial; Post-2021 focus.	The emerging data-driven systems being benchmarked against CAMS.
Meta-Model Features	**ERA5** Reanalysis (ECMWF)	0.25° spatial; Hourly; 2015–2024.	Provides meteorological predictors (wind, stability, shear) to train the failure-prediction meta-model.
Climate Projections	**CMIP6** (SSP2-4.5 & SSP5-8.5)	Monthly; 2030, 2040, 2050.	Used to generate synthetic future weather scenarios for testing.

3.3 Benchmarking Framework for Forecasting System Comparison

Systematic comparison of CAMS and selected AI-based systems relies on a verification framework that evaluates both deterministic accuracy and conditional performance across different atmospheric regimes.

3.3.1 Verification Metrics

Verification will utilize established atmospheric science metrics to ensure comparability with existing literature. The evaluation focuses on two dimensions: the general accuracy of the aerosol optical depth (AOD) magnitude and the ability to detect dust events correctly. The primary metrics are detailed in **Table 3.3**.

Table 3.3: Verification Metrics Framework

Category	Metric	Evaluation Purpose
Continuous	RMSE, Mean Bias, Correlation (r)	Quantifies the general magnitude of errors and systematic biases (e.g., does the model consistently over-/under-predict?).
Categorical	POD (Probability of Detection), FAR (False Alarm Ratio)	Assesses the model's ability to act as a binary warning system for dust events exceeding the 75th percentile.

Skill Scores	Fractional Gross Error (FGE)	Evaluates relative error performance, useful when comparing models with different resolutions.

3.3.2 Performance Stratification

To characterize under which conditions forecast typically fail, model errors will be analyzed against specific atmospheric and geographic drivers. This stratification strategy is summarized in Table 3.4.

Table 3.4: Performance Stratification Strategy

Stratification	Methodology	Objective
Seasonality	Standard Seasons (Winter, Spring, Summer, Autumn)	Assesses performance variability due to seasonal circulation shifts (e.g., Shamal vs. transitional seasons).
Source Direction	Wind Direction Analysis	Tests if errors are higher for specific wind sectors (e.g., NW flow from Iraq vs. S flow from the Empty Quarter).
Event Intensity	AOD Thresholds (Low <0.4, Moderate 0.4-0.8, High >0.8)	Determines if model reliability degrades during extreme events compared to background conditions.
Anthropogenic Influence	Trajectory Intersections	Investigate if air masses passing over known anthropogenic sources (dried marshes) correspond to higher forecast errors.

3.4 Meta-Model Development for Forecast Failure Prediction

The meta-modeling framework treats forecast error as a predictable variable. We formulate this as a binary classification problem: predicting whether a forecast will be "Reliable" or "Unreliable." A forecast is classified as unreliable if the normalized absolute error exceeds 1.5 times the climatological standard deviation.

3.4.1 Algorithm Selection

Three machine learning algorithms will be trained and compared:

1. Random Forest: Selected for robust performance on moderate datasets and built-in

feature importance.

2. XGBoost: A gradient boosting algorithm optimized for handling non-linear interactions in meteorological data.
3. Feedforward Neural Networks: A multi-layer perceptron architecture to capture complex dependencies.

3.4.2 Feature Engineering

Predictors are derived from ERA5 reanalysis at the time of forecast initialization. Key features include:

- Dynamic: Wind shear, boundary layer height, and vertical velocity.
- Thermodynamic: Stability indices, temperature gradients, and relative humidity.
- Temporal: Seasonality (sine/cosine encoding) and recent dust activity.

Model interpretability is ensured using SHAP (SHapley Additive exPlanations) values, which quantify the contribution of specific atmospheric conditions to the predicted probability of forecast failure.

3.5 Anthropogenic Signal Extraction Using Generative AI

A generative AI agent using advanced large language models will scan historical records for evidence of anthropogenic events that affect dust sources. The specific categories of source documents and the target information to be extracted are summarized in Table 3.2.

Table 3.2: Data Sources for Anthropogenic Event Extraction

Source Category	Specific Document Types	Target Information
Governmental Reports	Environmental assessments (UAE, Iraq, KSA), agricultural ministry reports, and water management records.	Land use changes, irrigation project failures, and official impact statements.
International Organizations	Publications from UNEP, FAO, and the World Bank.	Regional desertification trends, agricultural productivity assessments, and water scarcity studies.
Peer-Reviewed Literature	Academic journals and published case studies.	Analysis of specific dust source regions and conflict impact assessments.
News Archives	Regional news outlets (English and Arabic).	Real-time reporting on dust storms, agricultural collapse, and marsh drainage.

3.5.1 Validation and Model Integration

To ensure physical validity, AI-extracted events will be cross-referenced with satellite records: MODIS NDVI time-series will confirm vegetation loss for agricultural abandonment, and JRC Global Surface Water data will verify marsh drying. Confirmed events are then integrated into the meta-model as binary features whenever 48-hour back-trajectories indicate that air parcels transporting dust passed directly over these disturbed zones. This tests whether including dynamic anthropogenic sources improves failure prediction compared to static baselines.

3.6 Climate Robustness Assessment

To test model stability under non-stationary conditions, a Pseudo-Global Warming (PGW) approach will generate synthetic future meteorological datasets by superimposing CMIP6 climate anomalies (SSP2-4.5 and SSP5-8.5 for 2030–2050) onto historical ERA5 fields. The trained meta-models will be applied to these synthetic futures to estimate failure probabilities. We specifically test the hypothesis that AI-based models, lacking explicit physical constraints, will exhibit steeper performance degradation under climate-driven distribution shifts than physics-based systems such as CAMS.

3.7 Validation Strategy

Validity is ensured through strict temporal cross-validation (training on 2015–2020, testing on 2023–2024) to prevent data leakage. Meta-model skill will be benchmarked against persistence and climatology baselines, with spatial generalization tested via leave-one-station-out cross-validation. All performance differences between physics and AI approaches will be assessed using bootstrap resampling (1000 iterations) to generate 95% confidence intervals, ensuring that reported improvements are statistically significant.

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